STABLE PHASE RETRIEVAL FOR SPANS OF INDEPENDENT RANDOM VARIABLES

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Abstract. We study stable phase retrieval for closed spans of independent real-valued random variables in L^2. We prove the sufficiency direction of the Calderbank–Daubechies–Freeman–Freeman conjecture up to one exceptional coordinate: after L^2 normalization and relabeling, stable phase retrieval holds whenever all but possibly one coordinate satisfy a uniform two-sided L^1 bound. The paper gives two proofs. The first is a compactness proof based on a head–tail decomposition, bounded probe functions, and infinitely divisible limits. The second is a quantitative proof in the homogeneous case, using an explicit dichotomy and anticoncentration estimates for diffuse tails.

1. Introduction

The phase retrieval problem asks whether an unknown vector or function can be recovered from phaseless data. In the real setting, the unavoidable symmetry is multiplication by a global sign, and the central issue is whether this is the only obstruction. In finite dimensions, phase retrieval has by now a substantial theory, spanning injectivity, stability, convex relaxations, random measurements, and structured models [8, 25, 15, 14, 39, 24]. Infinite-dimensional phase retrieval is subtler. Recovery itself can hold in Hilbert spaces [12], but uniform stability can fail for continuous frames [2], and natural measurement systems such as Gabor families can be severely ill-posed [4, 35, 36, 46]. For general background we refer to [34]. Further landmarks include early reconstruction and finite-dimensional stability results [44, 43, 7, 8, 25, 15, 14, 39, 24, 38], variants for affine, vector-valued, and projection-based models [9, 20, 11, 10, 19, 1], and infinite-dimensional or structured-measurement results [13, 5, 21]. For the surrounding frame-theoretic and discretization background we also refer to [3, 17, 18, 16, 26, 28, 31, 32, 30, 37, 40, 23, 42, 6, 45, 41].

Two themes run through the subject. The first is uniqueness: does the map $f \mapsto |f|$ distinguish points of the model space modulo the natural symmetry? The second is stability: can one recover the signal in a uniformly continuous, or even Lipschitz, fashion from phaseless data? In applications, the second question is indispensable. Exact injectivity without stability is too weak for numerical reconstruction, and many infinite-dimensional models exhibit precisely this pathology. One of the main conceptual contributions of [29] is to make the stability question intrinsic to the geometry of the subspace rather than to a particular discretized frame. This point of view is particularly well suited to random subspaces of L^2, where the ambient structure is probabilistic from the outset and where orthogonality interacts naturally with independence. In a complementary direction, Christ,

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Pineau, and Taylor [22] constructed examples with only Hölder-stable recovery, underscoring that even when phase retrieval remains possible the quantitative modulus can deteriorate substantially.

The model considered here is also connected to the geometry of independent coordinates in classical Banach space theory. An independent sequence of centered L^2-normalized random variables behaves like a random coordinate system whose linear span carries both Euclidean and probabilistic structure. The size of the L^1 norm of each coordinate is a first quantitative measure of how much sign information survives after passing to absolute values. Variables with very small L^1 norm are too concentrated near zero, while variables with L^1 norm too close to one are nearly flat in modulus and hence almost insensitive to sign changes. The theorem proved in this paper says, in essence, that outside of those two obstructions there is enough randomness in the coordinate system to force uniform phase retrieval.

It is worth stressing that the problem is not coordinatewise. Stable phase retrieval asks for control of *all* vectors in the closed span, not merely for the generators. Thus one must understand how arbitrary linear combinations behave under the nonlinear map $f \mapsto |f|$. The difficult regime is when two orthogonal combinations have almost identical moduli. In that situation the coefficients may spread over infinitely many coordinates, and no finite-dimensional argument can be applied directly. The head–tail decomposition used below is precisely the mechanism that overcomes this: the few large coordinates form the head, which is rigid enough to be identified by bounded probes, while the many small coordinates form the tail, which is diffuse enough to admit probabilistic structure theorems.

We adopt the function-space viewpoint introduced in [29, 3]. Let X be a real function space and let $E \subset X$ be a subspace. We say that E does *stable phase retrieval* if there exists $C < \infty$ such that

$$\min_{\sigma \in \{\pm 1\}} \|f - \sigma g\|_X \leq C \, \||f| - |g|\|_X \qquad \text{for all } f, g \in E.$$

The theorem of Freeman, Oikhberg, Pineau, and Taylor [29] shows that in real Hilbert spaces the problem reduces to orthogonal pairs: to prove stable phase retrieval on a subspace it suffices to lower-bound the modulus gap for pairs of orthogonal unit vectors in that subspace. This orthogonal reduction is the common point of departure of both proofs given here. For historical continuity, we note that [27] is the preprint version of [29].

Our interest is the structure of subspaces generated by independent random variables. Calderbank, Daubechies, Freeman, and Freeman [13] proved stable phase retrieval for a shifted model and proposed the following conjectural characterization in the unshifted setting: if (ξ_i) is an orthonormal sequence of independent centered random variables in L^2, then $\overline{\mathrm{span}}(\xi_i)$ does stable phase retrieval if and only if there exist constants $0 < A \leq B < 1$ such that $A \leq \|\xi_i\|_{L^1} \leq B$ for all but at most one index. The lower bound excludes concentration near zero; the upper bound excludes the Rademacher regime in which $|\xi_i|$ is constant and therefore carries no sign information.

The phrase "at most one exceptional coordinate" is not an accident of the proof. It is the largest defect compatible with the expected geometry of the problem. One exceptional coordinate may coexist with an otherwise phase-retrieving family, but two genuinely exceptional coordinates already allow a nontrivial loss of sign information at the level of the span. The one-exception theorem should therefore be viewed as the natural endpoint of the qualitative theory.

The necessity of these conditions is by now standard. The equivalence of L^1 and L^2 norms on a phase-retrieving subspace follows from [29]; in particular a uniform positive lower L^1 bound is forced. Likewise, a centered L^2-normalized variable satisfies $\|\xi\|_{L^1} = 1$ if and only if $\xi^2 \equiv 1$, that is, if and only if ξ is Rademacher-valued. But a span containing two Rademacher directions cannot even do phase retrieval, since all such coordinates have the same absolute value.

Our first theorem gives the desired sufficiency result with one exceptional coordinate.

Theorem 1.1 (One-exception stable phase retrieval)**.** *Let $\eta, \xi_2, \xi_3, \dots$ be independent real-valued random variables and fix $\delta > 0$. Assume*

$$\|\eta\|_{L^2} = 1, \qquad \mathbb{E}[\eta] = 0,$$

and, for every $i \geq 2$,

$$\|\xi_i\|_{L^2} = 1, \qquad \mathbb{E}[\xi_i] = 0, \qquad \delta \leq \|\xi_i\|_{L^1} \leq 1 - \delta.$$

Then there exists $C_{\delta,\eta} \geq 1$ such that, for all $X, Y \in \overline{\mathrm{span}}(\eta, \xi_i : i \geq 2)$,

$$\min_{\sigma \in \{\pm 1\}} \|X - \sigma Y\|_{L^2} \leq C_{\delta,\eta} \, \||X| - |Y|\|_{L^2}.$$

Remark 1.2. After relabeling the coordinates and normalizing in L^2, Theorem 1.1 is exactly the sufficiency direction of the Calderbank–Daubechies–Freeman–Freeman conjecture up to one exceptional variable.

The second proof is quantitative in spirit. It keeps the one-exception framework of Theorem 1.1, but replaces the limit-law endpoint by an explicit head–tail dichotomy and by anticoncentration estimates for the diffuse part. In that sense, the compactness and quantitative proofs address the same theorem from complementary angles. The compactness proof shows that a bad sequence cannot exist by passing to a limit model; the quantitative proof shows directly that every orthogonal pair already contains a definite amount of modulus separation.

The two proofs illuminate complementary aspects of the theorem. The compactness proof begins with a normalized counterexample array and extracts a head–tail decomposition of the coefficients. The head consists of the few coordinates carrying macroscopic mass, while the tail is diffuse in the sense that its maximal coefficient goes to zero. Bounded probes read off quadratic coefficients from the observable $|X_n|^2 - |Y_n|^2$, forcing the head limit to agree up to sign. The tail then converges to an infinitely divisible law. Since the exact limit identity still has the form $|H_X + R_X| = |H_Y + R_Y|$, the support of the tail law is forced onto one of the phase-retrieval lines; tail rigidity then contradicts the orthogonal normalization. In the degenerate-head regime, the same contradiction can be reached by a different endpoint argument: diffuse tails produce a nontrivial Gaussian component in the limit law, and a line-geometry argument for convolutions supported on $\{(u, v) : |u| = |v|\}$ forces that Gaussian component, and hence the full law, onto a single line.

The quantitative proof replaces the limit law by an explicit dichotomy. The coordinates are split into a head and a tail according to a fixed cutoff. Either one of the signs $a \pm b$ is essentially supported on the head, in which case bounded probes recover the head difference matrix and orthogonality forces a positive modulus gap, or both $a + b$ and $a - b$ remain diffuse on the tail. In the latter regime one partitions the tail into many blocks of comparable energy and applies symmetrization,

Sperner-type combinatorics, and small-ball estimates to prove that both combinations stay away from zero with positive probability. This yields an explicit lower bound for $\mathbb{E}|X_a^2 - X_b^2|$ and hence for $\||X_a| - |X_b|\|_{L^2}$.

The paper is organized as follows. Section 2 collects the basic definitions and the analytic tools used throughout. In Section 3 we present the compactness proof of Theorem 1.1, including the Gaussian alternative to the final line-support argument. Section 4 gives a second proof of Theorem 1.1, explaining how the limit-law step can be bypassed by an explicit decomposition together with anticoncentration. Finally, Section 5 discusses extensions, related directions, and open problems. Although we now have a quantitative proof in the homogeneous setting, we have chosen to retain the compactness proof because of its flexibility: it isolates a robust mechanism that is likely to survive in more general symmetry classes and in settings where quantitative control is secondary.

2. Preliminaries

2.1. Stable phase retrieval and orthogonal reduction.

Definition 2.1. Let E be a subspace of a real Banach lattice X. We say that E does *stable phase retrieval* if there exists $C < \infty$ such that

$$\min_{\sigma \in \{\pm 1\}} \|f - \sigma g\|_X \le C \, \||f| - |g|\|_X \qquad (f, g \in E).$$

We shall use repeatedly the following orthogonal reduction from [29].

Proposition 2.2 (Orthogonal reduction)**.** *Let H be a real Hilbert space and let $E \subset H$ be a subspace. Then E does stable phase retrieval if and only if there exists $c > 0$ such that*

$$\||f| - |g|\|_H \ge c$$

for every orthogonal pair $f, g \in E$ with $\|f\|_H = \|g\|_H = 1$.

2.2. Probe functions and lower L^1 bounds.

The first input is the existence of bounded dual probes. These are the basic coefficient-extraction devices in both proofs.

Lemma 2.3 (Linear and quadratic probes)**.** *Let ξ be a real random variable with $\mathbb{E}[\xi] = 0$ and $\mathbb{E}[\xi^2] = 1$.*

(1) *If $\mathbb{E}[|\xi|] \ge A > 0$, then there exists a bounded measurable function S such that*

$$\mathbb{E}[S] = 0, \qquad \mathbb{E}[\xi S] = 1, \qquad \|S\|_{L^\infty} \le \frac{2}{A}.$$

(2) *If $\mathbb{E}[|\xi|] \le B < 1$, then there exists a bounded measurable function T such that*

$$\mathbb{E}[T] = 0, \qquad \mathbb{E}[(\xi^2 - 1)T] = 1, \qquad \|T\|_{L^\infty} \le \frac{2}{1 - B}.$$

Proof. We first construct the linear probe. Define

$$S_0 := \operatorname{sign}(\xi),$$

with the convention $\operatorname{sign}(0) = 0$. Then S_0 is measurable and satisfies $|S_0| \leq 1$ almost surely. Moreover,

$$\xi S_0 = |\xi| \qquad \text{almost surely,}$$

so

$$\mathbb{E}[\xi S_0] = \mathbb{E}[|\xi|] \geq A > 0.$$

This positivity allows us to normalize. Set

$$S := \frac{S_0 - \mathbb{E}[S_0]}{\mathbb{E}[\xi S_0]}.$$

The function S is measurable and bounded. Its mean is

$$\mathbb{E}[S] = \frac{\mathbb{E}[S_0] - \mathbb{E}[S_0]}{\mathbb{E}[\xi S_0]} = 0,$$

and, using that $\mathbb{E}[\xi] = 0$,

$$\mathbb{E}[\xi S] = \frac{\mathbb{E}[\xi S_0] - \mathbb{E}[S_0]\mathbb{E}[\xi]}{\mathbb{E}[\xi S_0]} = \frac{\mathbb{E}[\xi S_0]}{\mathbb{E}[\xi S_0]} = 1.$$

Finally, since $|S_0| \leq 1$ and $|\mathbb{E}[S_0]| \leq \mathbb{E}|S_0| \leq 1$,

$$|S_0 - \mathbb{E}[S_0]| \leq 2 \qquad \text{almost surely,}$$

and therefore

$$\|S\|_{L^\infty} \leq \frac{2}{\mathbb{E}[\xi S_0]} = \frac{2}{\mathbb{E}[|\xi|]} \leq \frac{2}{A}.$$

This proves part (1).

For the quadratic probe, apply the same idea to the centered variable

$$\eta := \xi^2 - 1.$$

Because $\mathbb{E}[\xi^2] = 1$, one has $\mathbb{E}[\eta] = 0$. We now show that η has positive L^1 mass bounded away from zero. Since $\xi^2 = |\xi|^2$,

$$|\eta| = ||\xi|^2 - 1| = ||\xi| - 1|(|\xi| + 1) \geq ||\xi| - 1|.$$

Taking expectations and using the reverse triangle inequality yields

$$\mathbb{E}[|\eta|] \geq \mathbb{E}[||\xi| - 1|] \geq \big|\mathbb{E}[|\xi|] - 1\big| = 1 - \mathbb{E}[|\xi|] \geq 1 - B.$$

Now set

$$T_0 := \operatorname{sign}(\eta)$$

and define

$$T := \frac{T_0 - \mathbb{E}[T_0]}{\mathbb{E}[\eta T_0]} = \frac{T_0 - \mathbb{E}[T_0]}{\mathbb{E}[|\eta|]}.$$

Exactly as in the linear case, T is measurable, bounded, and satisfies

$$\mathbb{E}[T] = 0, \qquad \mathbb{E}[\eta T] = 1.$$

Since $\eta = \xi^2 - 1$, this reads

$$\mathbb{E}[(\xi^2 - 1)T] = 1.$$

The L^∞ bound is

$$\|T\|_{L^\infty} \le \frac{2}{\mathbb{E}[|\eta|]} \le \frac{2}{1-B}.$$

This completes the proof. □

Remark 2.4. If $\xi_1, \dots, \xi_n$ are independent centered random variables and $X_c = \sum_i c_i \xi_i$, then the centering of the probes eliminates every coordinate except the one being tested. In particular,

$$\mathbb{E}[X_c\, S_i] = c_i, \qquad \mathbb{E}[X_c^2\, T_i] = c_i^2, \qquad \mathbb{E}[X_c^2\, S_i S_j] = 2c_i c_j \quad (i \ne j),$$

whenever S_i, S_j, T_i are the corresponding probes for ξ_i, ξ_j.

The second basic estimate is the lower L^1 bound for independent sums.

Lemma 2.5 (L^1 lower bound for independent sums)**.** *Let $\zeta_1, \dots, \zeta_N$ be independent centered real random variables and assume that*

$$\|\zeta_i\|_{L^1} \ge \delta \|\zeta_i\|_{L^2} \qquad (1 \le i \le N).$$

Then for every scalar family $(c_i)_{i=1}^N$,

$$\|\sum_{i=1}^N c_i \zeta_i\|_{L^1} \ge \frac{\delta}{2\sqrt{2}} \Big(\sum_{i=1}^N c_i^2 \|\zeta_i\|_{L^2}^2 \Big)^{1/2}.$$

Proof. The statement is homogeneous in the coefficient vector, so we may first normalize and assume that

$$\sum_{i=1}^N c_i^2 \|\zeta_i\|_{L^2}^2 = 1.$$

Indeed, once the claim is established under this normalization, the general case follows by scaling.

We next reduce to the unit-variance case. For each i set

$$\widetilde{\zeta}_i := \frac{\zeta_i}{\|\zeta_i\|_{L^2}}, \qquad \widetilde{c}_i := c_i \|\zeta_i\|_{L^2}.$$

Then the variables $\widetilde{\zeta}_i$ are still independent and centered, they satisfy $\|\widetilde{\zeta}_i\|_{L^2} = 1$, and the lower L^1 bound becomes

$$\|\widetilde{\zeta}_i\|_{L^1} = \frac{\|\zeta_i\|_{L^1}}{\|\zeta_i\|_{L^2}} \ge \delta.$$

Moreover,

$$\sum_{i=1}^N \widetilde{c}_i \widetilde{\zeta}_i = \sum_{i=1}^N c_i \zeta_i, \qquad \sum_{i=1}^N \widetilde{c}_i^2 = 1.$$

Thus it is enough to prove the lemma under the additional assumptions

$$\|\zeta_i\|_{L^2} = 1 \qquad (1 \le i \le N), \qquad \sum_{i=1}^N c_i^2 = 1.$$

Define

$$S := \sum_{i=1}^N c_i \zeta_i,$$

and let

$$S' := \sum_{i=1}^{N} c_i \zeta_i'$$

be an independent copy of S, obtained from an independent copy $(\zeta_i')_{i=1}^N$ of the family $(\zeta_i)_{i=1}^N$. By the triangle inequality,

$$|S - S'| \le |S| + |S'|.$$

Taking expectations and using that S and S' have the same distribution, we obtain

$$\|S - S'\|_{L^1} \le 2\|S\|_{L^1}.$$

Equivalently,

$$2\|S\|_{L^1} \ge \|S - S'\|_{L^1}.$$

Now the random variables $\zeta_i - \zeta_i'$ are independent and symmetric. Conditioning on their absolute values and applying the Khintchine inequality gives

$$\|S - S'\|_{L^1} = \Big\|\sum_{i=1}^{N} c_i(\zeta_i - \zeta_i')\Big\|_{L^1} \ge \frac{1}{\sqrt{2}}\,\mathbb{E}\Big[\Big(\sum_{i=1}^{N} c_i^2(\zeta_i - \zeta_i')^2\Big)^{1/2}\Big].$$

Because the nonnegative numbers $(c_i^2)_{i=1}^N$ sum to one, Cauchy–Schwarz implies the pointwise estimate

$$\sum_{i=1}^{N} c_i^2|\zeta_i - \zeta_i'| \le \Big(\sum_{i=1}^{N} c_i^2\Big)^{1/2}\Big(\sum_{i=1}^{N} c_i^2(\zeta_i - \zeta_i')^2\Big)^{1/2} = \Big(\sum_{i=1}^{N} c_i^2(\zeta_i - \zeta_i')^2\Big)^{1/2}.$$

Taking expectations and combining with the previous inequality, we arrive at

$$\|S - S'\|_{L^1} \ge \frac{1}{\sqrt{2}} \sum_{i=1}^{N} c_i^2 \|\zeta_i - \zeta_i'\|_{L^1}.$$

Hence

$$\|S\|_{L^1} \ge \frac{1}{2\sqrt{2}} \sum_{i=1}^{N} c_i^2 \|\zeta_i - \zeta_i'\|_{L^1}.$$

It remains to bound each $\|\zeta_i - \zeta_i'\|_{L^1}$ from below. Since ζ_i' is independent of ζ_i and centered,

$$\mathbb{E}[\zeta_i - \zeta_i' \mid \zeta_i] = \zeta_i.$$

Applying Jensen's inequality to the conditional expectation gives

$$\|\zeta_i - \zeta_i'\|_{L^1} \ge \|\mathbb{E}[\zeta_i - \zeta_i' \mid \zeta_i]\|_{L^1} = \|\zeta_i\|_{L^1} \ge \delta.$$

Substituting this into the previous bound yields

$$\|S\|_{L^1} \ge \frac{\delta}{2\sqrt{2}} \sum_{i=1}^{N} c_i^2 = \frac{\delta}{2\sqrt{2}}.$$

Since this is exactly the normalized form of the desired estimate, the proof is complete. □

2.3. **Diffusive arrays and infinitely divisible limits.** We shall use the classical fact that weak limits of infinitesimal triangular arrays of independent random variables are infinitely divisible; see, for instance, [33, Chapter IV].

Lemma 2.6 (Diffuse tail rigidity)**.** *Let $(\xi_{n,i})_{n\geq 1,\ i\geq 2}$ be a triangular array such that for each fixed n the family $(\xi_{n,i})_{i\geq 2}$ is independent, centered, L^2-normalized, and satisfies*

$$\|\xi_{n,i}\|_{L^1} \geq \delta \qquad (i \geq 2).$$

Let $c^{(n)} \in \ell^2(\mathbb{N})$ satisfy

$$\sup_n \|c^{(n)}\|_2 < \infty, \qquad c_1^{(n)} = 0, \qquad \max_{i\geq 2} |c_i^{(n)}| \to 0.$$

Set

$$Z_n := \sum_{i\geq 2} c_i^{(n)} \xi_{n,i}.$$

If $Z_n \to 0$ in distribution, then $\|c^{(n)}\|_2 \to 0$.

Proof. Because the limiting law is the Dirac mass at 0, the assumption $Z_n \to 0$ in distribution implies that

$$Z_n \to 0 \qquad \text{in probability.}$$

To pass from convergence in probability to convergence of first moments, we use uniform integrability.

For each n, the variables $(\xi_{n,i})_{i\geq 2}$ are independent, centered, and L^2-normalized. Therefore they form an orthonormal family in L^2, and the random sum Z_n has

$$\|Z_n\|_{L^2}^2 = \mathbb{E}\Big[\Big(\sum_{i\geq 2} c_i^{(n)} \xi_{n,i}\Big)^2\Big] = \sum_{i\geq 2} (c_i^{(n)})^2 \mathbb{E}[\xi_{n,i}^2] = \sum_{i\geq 2} (c_i^{(n)})^2 = \|c^{(n)}\|_2^2.$$

Since $\sup_n \|c^{(n)}\|_2 < \infty$, the sequence $(Z_n)_n$ is bounded in L^2. Any L^2-bounded family is uniformly integrable in L^1, so $(|Z_n|)_n$ is uniformly integrable. Because $Z_n \to 0$ in probability, we may therefore conclude that

$$\|Z_n\|_{L^1} \longrightarrow 0.$$

Now apply Lemma 2.5 to each row of the array. The assumptions of that lemma are satisfied because every $\xi_{n,i}$ is centered, L^2-normalized, and obeys $\|\xi_{n,i}\|_{L^1} \geq \delta$. Hence

$$\|Z_n\|_{L^1} = \Big\|\sum_{i\geq 2} c_i^{(n)} \xi_{n,i}\Big\|_{L^1} \geq \frac{\delta}{2\sqrt{2}} \Big(\sum_{i\geq 2} (c_i^{(n)})^2\Big)^{1/2} = \frac{\delta}{2\sqrt{2}} \|c^{(n)}\|_2.$$

The left-hand side tends to zero, so the right-hand side must also tend to zero. Therefore

$$\|c^{(n)}\|_2 \longrightarrow 0,$$

which is the desired conclusion. □

3. First Proof: Compactness

This section proves Theorem 1.1. We follow the compactness route from the manuscript 'compactness_formal.tex', but reorganize it so that the head–tail structure and the endpoint alternatives are explicit.

3.1. Contradiction array and profile decomposition.

Proposition 3.1 (Normalized counterexample array)**.** *If Theorem 1.1 fails, then there exist*

(i) *a triangular array* $(\xi_{n,i})_{n\geq 1,\ i\geq 2}$ *such that for each fixed* n *the family* $(\eta, \xi_{n,i})_{i\geq 2}$ *is independent and every* $\xi_{n,i}$ *satisfies*

$$\|\xi_{n,i}\|_{L^2} = 1, \qquad \mathbb{E}[\xi_{n,i}] = 0, \qquad \delta \leq \|\xi_{n,i}\|_{L^1} \leq 1 - \delta;$$

(ii) *coefficient vectors* $a^{(n)}, b^{(n)} \in \ell^2(\mathbb{N})$ *with*

$$\|a^{(n)}\|_2 = \|b^{(n)}\|_2 = 1, \qquad \langle a^{(n)}, b^{(n)} \rangle = 0,$$

such that, writing

$$X_n := a_1^{(n)}\eta + \sum_{i\geq 2} a_i^{(n)}\xi_{n,i}, \qquad Y_n := b_1^{(n)}\eta + \sum_{i\geq 2} b_i^{(n)}\xi_{n,i},$$

one has

$$\||X_n| - |Y_n|\|_{L^2} \longrightarrow 0.$$

Proof. Assume that the conclusion of Theorem 1.1 is false. Then there is no constant C for which the stable phase retrieval estimate holds uniformly on the closed span of the coordinates. By the orthogonal reduction (Proposition 2.2), this means that we may find, for each $n \in \mathbb{N}$, an orthogonal pair of unit vectors in the span whose modulus gap is at most $1/n$. Concretely, there exist functions X_n, Y_n in $\overline{\mathrm{span}}(\eta, \xi_i : i \geq 2)$ such that

$$\|X_n\|_{L^2} = \|Y_n\|_{L^2} = 1, \qquad \mathbb{E}[X_n Y_n] = 0, \qquad \||X_n| - |Y_n|\|_{L^2} \leq \frac{1}{n}.$$

Because the family $(\eta, \xi_i)_{i\geq 2}$ is orthonormal in L^2, we may write

$$X_n = a_1^{(n)}\eta + \sum_{i\geq 2} a_i^{(n)}\xi_i, \qquad Y_n = b_1^{(n)}\eta + \sum_{i\geq 2} b_i^{(n)}\xi_i,$$

with coefficient vectors $a^{(n)}, b^{(n)} \in \ell^2(\mathbb{N})$. Parseval's identity then yields

$$\|a^{(n)}\|_2 = \|X_n\|_{L^2} = 1, \qquad \|b^{(n)}\|_2 = \|Y_n\|_{L^2} = 1,$$

and, since the two functions are orthogonal in L^2,

$$\langle a^{(n)}, b^{(n)} \rangle = \mathbb{E}[X_n Y_n] = 0.$$

Finally, to prepare for later rowwise permutations of the good coordinates, we rename the good variables used in the n-th row by writing them as $(\xi_{n,i})_{i\geq 2}$. This does not alter their laws or any of the stated hypotheses, but it is convenient for the compactness scheme. The displayed modulus convergence is exactly our chosen violating sequence. This proves the proposition. □

Lemma 3.2 (Reordered profile decomposition)**.** *After passing to a subsequence and, row by row, applying the same permutation to the good coordinates and to their coefficients, one may find vectors* $a, b \in \ell^2(\mathbb{N})$ *and integers* $m_n \to \infty$ *such that, with*

$$a_{\mathrm{head}}^{(n)} := a^{(n)}\mathbf{1}_{\{1,\dots,m_n\}}, \qquad a_{\mathrm{tail}}^{(n)} := a^{(n)}\mathbf{1}_{\{m_n+1, m_n+2, \dots\}},$$

and likewise for $b^{(n)}$*, the following hold:*

(1) $a_{\mathrm{head}}^{(n)} \to a$ *and* $b_{\mathrm{head}}^{(n)} \to b$ *strongly in* ℓ^2*;*

(2)
$$\max_{i>m_n} \max(|a_i^{(n)}|, |b_i^{(n)}|) \to 0;$$

(3)
$$\|a_{\text{tail}}^{(n)}\|_2^2 \to 1 - \|a\|_2^2, \qquad \|b_{\text{tail}}^{(n)}\|_2^2 \to 1 - \|b\|_2^2,$$

and
$$\langle a_{\text{tail}}^{(n)}, b_{\text{tail}}^{(n)} \rangle \to -\langle a, b \rangle.$$

Proof. For each row n and each good index $i \geq 2$, define
$$s_i^{(n)} := \max\big(|a_i^{(n)}|, |b_i^{(n)}|\big).$$
Permuting the good coordinates within the n-th row, we may assume that the sequence $(s_i^{(n)})_{i\geq 2}$ is nonincreasing in i. The same permutation is applied simultaneously to the variables $\xi_{n,i}$ and to both coefficient vectors, so no hypothesis is changed.

The monotone rearrangement gives a uniform pointwise bound on the reordered coefficients. Indeed,
$$\sum_{i\geq 2} (s_i^{(n)})^2 \leq \sum_{i\geq 2} \big((a_i^{(n)})^2 + (b_i^{(n)})^2\big) \leq 2.$$
Since $(s_i^{(n)})_{i\geq 2}$ is nonincreasing, for every $i \geq 2$ we have
$$(i-1)(s_i^{(n)})^2 \leq \sum_{j=2}^{i} (s_j^{(n)})^2 \leq 2,$$
and therefore
$$s_i^{(n)} \leq \sqrt{\frac{2}{i-1}}.$$
In particular, for each fixed coordinate i, the sequences $\big(a_i^{(n)}\big)_{n\geq 1}$ and $\big(b_i^{(n)}\big)_{n\geq 1}$ are bounded.

We now extract coordinatewise limits. First pass to a subsequence so that $a_1^{(n)}$ and $b_1^{(n)}$ converge. Next pass to a further subsequence so that $a_2^{(n)}$ and $b_2^{(n)}$ converge, and continue inductively. A standard diagonal argument yields a single subsequence, not relabeled, for which every fixed coordinate converges:
$$a_i^{(n)} \to a_i, \qquad b_i^{(n)} \to b_i \qquad (i \geq 1).$$
Set
$$a := (a_i)_{i\geq 1}, \qquad b := (b_i)_{i\geq 1}.$$

We claim that $a, b \in \ell^2(\mathbb{N})$. Fix $M \in \mathbb{N}$. Since the first M coordinates converge,
$$\sum_{i=1}^{M} a_i^2 = \lim_{n\to\infty} \sum_{i=1}^{M} (a_i^{(n)})^2 \leq \liminf_{n\to\infty} \sum_{i\geq 1} (a_i^{(n)})^2 = 1.$$
Letting $M \to \infty$ and using monotone convergence gives $\sum_i a_i^2 \leq 1$. The same argument yields $\sum_i b_i^2 \leq 1$.

Because $a, b \in \ell^2$, we may choose an increasing sequence of integers $(m_k)_{k\geq 1}$ with $m_k \to \infty$ such that
$$\sum_{i>m_k} a_i^2 + \sum_{i>m_k} b_i^2 \leq 2^{-k} \qquad (k \geq 1).$$

For each fixed k, coordinatewise convergence implies

$$\sum_{i=1}^{m_k}|a_i^{(n)}-a_i|^2+\sum_{i=1}^{m_k}|b_i^{(n)}-b_i|^2\longrightarrow 0 \qquad (n\to\infty).$$

Passing to another subsequence if necessary, we may assume that for each n,

$$\sum_{i=1}^{m_n}|a_i^{(n)}-a_i|^2+\sum_{i=1}^{m_n}|b_i^{(n)}-b_i|^2\le 2^{-n}.$$

We now verify the three conclusions. First,

$$\|a_{\text{head}}^{(n)}-a\|_2^2\le\sum_{i=1}^{m_n}|a_i^{(n)}-a_i|^2+\sum_{i>m_n}a_i^2\le 2^{-n}+2^{-n},$$

so $a_{\text{head}}^{(n)}\to a$ in ℓ^2. The same estimate gives $b_{\text{head}}^{(n)}\to b$.

Second, if $i>m_n$, then by monotonicity,

$$\max\bigl(|a_i^{(n)}|,|b_i^{(n)}|\bigr)=s_i^{(n)}\le s_{m_n+1}^{(n)}\le\sqrt{\frac{2}{m_n}}.$$

Since $m_n\to\infty$, this proves

$$\max_{i>m_n}\max\bigl(|a_i^{(n)}|,|b_i^{(n)}|\bigr)\to 0.$$

Finally, because each full coefficient vector has ℓ^2 norm one,

$$\|a_{\text{tail}}^{(n)}\|_2^2=1-\|a_{\text{head}}^{(n)}\|_2^2\longrightarrow 1-\|a\|_2^2,$$

and similarly

$$\|b_{\text{tail}}^{(n)}\|_2^2\longrightarrow 1-\|b\|_2^2.$$

For the inner products, use the orthogonality of the full vectors:

$$0=\langle a^{(n)},b^{(n)}\rangle=\langle a_{\text{head}}^{(n)},b_{\text{head}}^{(n)}\rangle+\langle a_{\text{tail}}^{(n)},b_{\text{tail}}^{(n)}\rangle.$$

Since the head pieces converge strongly,

$$\langle a_{\text{head}}^{(n)},b_{\text{head}}^{(n)}\rangle\to\langle a,b\rangle,$$

and therefore

$$\langle a_{\text{tail}}^{(n)},b_{\text{tail}}^{(n)}\rangle\to-\langle a,b\rangle.$$

This proves the lemma. □

3.2. **Head identification.**

Proposition 3.3 (Exceptional probes)**.** *There exists a bounded measurable function S_η such that*

$$\mathbb{E}[S_\eta]=0,\qquad \mathbb{E}[\eta S_\eta]=1.$$

If $\eta^2\not\equiv 1$ almost surely, then there exists a bounded measurable function T_η such that

$$\mathbb{E}[T_\eta]=0,\qquad \mathbb{E}[(\eta^2-1)T_\eta]=1.$$

Proof. Since $\|\eta\|_{L^2}=1$, the random variable η is not almost surely zero. Hence $\mathbb{E}[|\eta|]>0$. Applying the linear-probe construction from Lemma 2.3 with $A=\mathbb{E}[|\eta|]$ produces a bounded measurable

function S_η satisfying

$$\mathbb{E}[S_\eta] = 0, \qquad \mathbb{E}[\eta S_\eta] = 1.$$

Assume now that $\eta^2 \not\equiv 1$ almost surely. Then the centered random variable $\eta^2 - 1$ is not almost surely zero, because

$$\mathbb{E}[\eta^2 - 1] = \mathbb{E}[\eta^2] - 1 = 0.$$

Therefore its L^1 norm is strictly positive:

$$\mathbb{E}[|\eta^2 - 1|] > 0.$$

Applying again the linear-probe construction, now to the random variable $\eta^2 - 1$, we obtain a bounded measurable function T_η such that

$$\mathbb{E}[T_\eta] = 0, \qquad \mathbb{E}[(\eta^2 - 1)T_\eta] = 1.$$

This is exactly the desired conclusion. □

Lemma 3.4 (Identification of the head profile)**.** *Either there exists $\tau \in \{\pm 1\}$ such that*

$$a = \tau b \qquad \textit{and} \qquad \|a^{(n)}_{\text{head}} - \tau b^{(n)}_{\text{head}}\|_2 \to 0,$$

or else η is a Rademacher variable and

$$a_i = b_i = 0 \qquad (i \geq 2).$$

Proof. We first convert the convergence of moduli into convergence of the quadratic observable. Since

$$\|X_n\|_{L^2} = \|Y_n\|_{L^2} = 1,$$

we have

$$\||X_n| + |Y_n|\|_{L^2} \leq 2.$$

Therefore

$$\||X_n|^2 - |Y_n|^2\|_{L^1} = \|(|X_n| - |Y_n|)(|X_n| + |Y_n|)\|_{L^1} \leq 2\||X_n| - |Y_n|\|_{L^2} \longrightarrow 0.$$

Because the variables are real-valued, $|X_n|^2 = X_n^2$ and $|Y_n|^2 = Y_n^2$, so we have established

$$\|X_n^2 - Y_n^2\|_{L^1} \longrightarrow 0.$$

Step 1: diagonal good-coordinate identities. Fix $i \geq 2$ and let $T_{n,i}$ be the quadratic probe associated with the coordinate $\xi_{n,i}$. The probes are uniformly bounded in L^∞, so the L^1 convergence above implies

$$\mathbb{E}[(X_n^2 - Y_n^2)T_{n,i}] \longrightarrow 0.$$

We compute the left-hand side explicitly. Expanding X_n^2 and integrating against $T_{n,i}$, every term involving a coordinate different from $\xi_{n,i}$ vanishes because $T_{n,i}$ depends only on $\xi_{n,i}$ and is centered. Mixed terms involving $\xi_{n,i}$ and another coordinate also vanish by independence and centering. The only surviving contribution is the term $(a_i^{(n)})^2\xi_{n,i}^2$. Hence

$$\mathbb{E}[X_n^2 T_{n,i}] = (a_i^{(n)})^2\mathbb{E}[\xi_{n,i}^2 T_{n,i}] = (a_i^{(n)})^2,$$

because

$$\mathbb{E}[\xi_{n,i}^2 T_{n,i}] = \mathbb{E}[(\xi_{n,i}^2 - 1)T_{n,i}] + \mathbb{E}[T_{n,i}] = 1 + 0.$$

Similarly,

$$\mathbb{E}[Y_n^2 T_{n,i}] = (b_i^{(n)})^2.$$

Therefore

$$(a_i^{(n)})^2 - (b_i^{(n)})^2 = \mathbb{E}[(X_n^2 - Y_n^2)T_{n,i}] \longrightarrow 0.$$

Passing to the limit in n gives

$$a_i^2 = b_i^2 \qquad (i \geq 2).$$

Step 2: off-diagonal good-good identities. Fix distinct indices $i, j \geq 2$ and let $S_{n,i}, S_{n,j}$ be the corresponding linear probes. Again boundedness and L^1 convergence imply

$$\mathbb{E}[(X_n^2 - Y_n^2)S_{n,i}S_{n,j}] \longrightarrow 0.$$

In the expansion of X_n^2, all terms vanish after integration against $S_{n,i}S_{n,j}$ except the mixed term $2a_i^{(n)}a_j^{(n)}\xi_{n,i}\xi_{n,j}$. Indeed, any term missing one of the coordinates $\xi_{n,i}$ or $\xi_{n,j}$ is killed by the zero mean of the corresponding probe, and terms involving other coordinates factor through zero means by independence. Hence

$$\mathbb{E}[X_n^2 S_{n,i}S_{n,j}] = 2a_i^{(n)}a_j^{(n)}\mathbb{E}[\xi_{n,i}S_{n,i}]\,\mathbb{E}[\xi_{n,j}S_{n,j}] = 2a_i^{(n)}a_j^{(n)}.$$

Similarly,

$$\mathbb{E}[Y_n^2 S_{n,i}S_{n,j}] = 2b_i^{(n)}b_j^{(n)}.$$

Thus

$$2(a_i^{(n)}a_j^{(n)} - b_i^{(n)}b_j^{(n)}) \longrightarrow 0,$$

and passing to the limit gives

$$a_i a_j = b_i b_j \qquad (i \neq j,\ i, j \geq 2).$$

Step 3: exceptional-good identities. Fix $j \geq 2$. Let S_η be the bounded probe from Proposition 3.3. By the same reasoning,

$$\mathbb{E}[(X_n^2 - Y_n^2)S_\eta S_{n,j}] \longrightarrow 0.$$

Once more only one mixed term survives, namely $2a_1^{(n)}a_j^{(n)}\eta\xi_{n,j}$ for X_n^2 and the corresponding term for Y_n^2. Therefore

$$\mathbb{E}[X_n^2 S_\eta S_{n,j}] = 2a_1^{(n)}a_j^{(n)}\mathbb{E}[\eta S_\eta]\mathbb{E}[\xi_{n,j}S_{n,j}] = 2a_1^{(n)}a_j^{(n)},$$

and analogously

$$\mathbb{E}[Y_n^2 S_\eta S_{n,j}] = 2b_1^{(n)}b_j^{(n)}.$$

Hence

$$a_1 a_j = b_1 b_j \qquad (j \geq 2).$$

Step 4: deduction of the sign. Assume first that there exists some index $i_0 \geq 2$ such that $a_{i_0} \neq 0$. From the diagonal identity we know that $a_{i_0}^2 = b_{i_0}^2$, so $b_{i_0} = \tau a_{i_0}$ for some $\tau \in \{\pm 1\}$. Then for every $j \geq 2$, the off-diagonal identity gives

$$a_{i_0}a_j = b_{i_0}b_j = \tau a_{i_0}b_j.$$

Since $a_{i_0} \neq 0$, we conclude that $a_j = \tau b_j$ for every $j \geq 2$. Applying the exceptional-good identity with $j = i_0$, we obtain

$$a_1 a_{i_0} = b_1 b_{i_0} = \tau a_{i_0} b_1,$$

and again $a_{i_0} \neq 0$ implies $a_1 = \tau b_1$. Hence $a = \tau b$.

Assume next that

$$a_i = 0 \qquad \text{for every } i \geq 2.$$

Then the diagonal identities show that also $b_i = 0$ for every $i \geq 2$. If $\eta^2 \not\equiv 1$ almost surely, let T_η be the quadratic probe from Proposition 3.3. Repeating the diagonal argument for the exceptional coordinate gives

$$(a_1^{(n)})^2 - (b_1^{(n)})^2 = \mathbb{E}[(X_n^2 - Y_n^2)T_\eta] \longrightarrow 0,$$

whence $a_1^2 = b_1^2$ and therefore $a = \tau b$ for some sign τ.

The only remaining possibility is that $\eta^2 \equiv 1$ almost surely. Since $\mathbb{E}[\eta] = 0$ and $\|\eta\|_{L^2} = 1$, this means exactly that η is a Rademacher variable. In that case we are in the second alternative stated in the lemma.

Finally, if the first alternative holds, then the convergence of the head profiles follows from Lemma 3.2:

$$\|a_{\text{head}}^{(n)} - \tau b_{\text{head}}^{(n)}\|_2 \leq \|a_{\text{head}}^{(n)} - a\|_2 + \|b_{\text{head}}^{(n)} - b\|_2 \longrightarrow 0.$$

□

3.3. **Limit law and exact modulus identity.**

Lemma 3.5 (Limit law)**.** *After passing to a subsequence, the quadruple*

$$\left(X_n^{\text{head}}, Y_n^{\text{head}}, X_n^{\text{tail}}, Y_n^{\text{tail}}\right)$$

converges in distribution to a random vector

$$(H_X, H_Y, R_X, R_Y) \in \mathbb{R}^4$$

such that:

(1) (H_X, H_Y) *is independent of* (R_X, R_Y)*;*
(2) *the law of* (R_X, R_Y) *is infinitely divisible;*
(3) *either* $H_X = \tau H_Y$ *almost surely for some* $\tau \in \{\pm 1\}$*, or else* η *is Rademacher and* $H_X = a_1\eta$*,* $H_Y = b_1\eta$ *almost surely;*
(4)
$$|H_X + R_X| = |H_Y + R_Y| \qquad \textit{almost surely.}$$

Proof. The head variables converge in L^2 by the strong convergence of the head coefficients and the orthonormality of each row. The tail coefficient arrays are infinitesimal by Lemma 3.2, so every weak limit of the tail pair is infinitely divisible. Independence of head and tail persists because they are built from disjoint independent coordinates. Finally, $\||X_n| - |Y_n|\|_{L^2} \to 0$ implies $|X_n| - |Y_n| \to 0$ in probability; passing to the joint weak limit gives the exact identity in the limit. The description of the head follows from Lemma 3.4. □

Lemma 3.6 (Infinitely divisible laws on the phase-retrieval cone are line-supported)**.** *Let μ be an infinitely divisible probability measure on $\mathbb{R}^2$ with*

$$\operatorname{supp}\mu \subset \Gamma := \{(u,v) \in \mathbb{R}^2 : |u| = |v|\}.$$

Then there exists $\tau \in \{\pm 1\}$ such that

$$\operatorname{supp}\mu \subset L_\tau := \{(u, \tau u) : u \in \mathbb{R}\}.$$

Proof. The set Γ is exactly the union $L_+ \cup L_-$. Since μ is infinitely divisible, it is in particular 2-divisible: there exists a probability measure ν such that $\mu = \nu * \nu$. Let $A = \operatorname{supp}\nu$. Then

$$\operatorname{supp}\mu = \overline{A + A} \subset L_+ \cup L_-.$$

If $x, y \in A$ are distinct, then $2x, x+y, 2y \in A + A$. If $2x$ and $2y$ belong to the same line L_τ, then so does $x + y$, forcing $x, y \in L_\tau$. If they belong to different lines, then the midpoint $x + y$ is the origin, so $y = -x$ and again x, y lie in one of the lines. Thus every two points of A lie on a common phase-retrieval line. Since $L_+ \cap L_- = \{0\}$, this implies that A is contained in one of the two lines, and so is $\operatorname{supp}\mu$. □

3.4. **End of the compactness proof.**

Proof of Theorem 1.1. Assume that Theorem 1.1 fails and let the counterexample array be as in Proposition 3.1. Apply Lemmas 3.2, 3.4, and 3.5.

If the first alternative of Lemma 3.4 holds, fix $\tau \in \{\pm 1\}$ such that $H_Y = \tau H_X$ almost surely. The exact modulus identity becomes

$$|H_X + R_X| = |\tau H_X + R_Y| \qquad \text{almost surely.}$$

If H_X is not almost surely zero, then conditioning on the tail and varying the head over two support points of H_X shows that $R_Y - \tau R_X = 0$ almost surely. If $H_X \equiv 0$, then (R_X, R_Y) itself satisfies $|R_X| = |R_Y|$ almost surely, so Lemma 3.6 yields $R_Y = \tau_0 R_X$ almost surely for some $\tau_0 \in \{\pm 1\}$.

In either case, for a suitable sign $\sigma \in \{\pm 1\}$ the random variables

$$Z_n := \sum_{i > m_n} (b_i^{(n)} - \sigma a_i^{(n)}) \xi_{n,i}$$

converge in distribution to 0. Since the coefficients are diffuse, Lemma 2.6 gives

$$\|b_{\text{tail}}^{(n)} - \sigma a_{\text{tail}}^{(n)}\|_2 \to 0.$$

Together with Lemma 3.4 this implies $\|b^{(n)} - \sigma a^{(n)}\|_2 \to 0$, contradicting

$$\|b^{(n)} - \sigma a^{(n)}\|_2^2 = \|a^{(n)}\|_2^2 + \|b^{(n)}\|_2^2 - 2\sigma \langle a^{(n)}, b^{(n)} \rangle = 2.$$

If the second alternative of Lemma 3.4 holds, then η is Rademacher and the head limits are multiples of η alone. The same argument applies, since in that case the good coordinates disappear from the head and the tail limit still lies on Γ. □

3.5. **A Gaussian alternative in the degenerate-head regime.** The previous endpoint used only the support structure of infinitely divisible limits. The next result records a second route,

adapted from the complex one-exception manuscript, showing that in the degenerate-head regime the diffuse tail automatically generates a nontrivial Gaussian component.

Proposition 3.7 (Diffuse tails force a nontrivial Gaussian part)**.** *Let*

$$V_{n,i} := (a_i^{(n)}\xi_{n,i},\, b_i^{(n)}\xi_{n,i}) \in \mathbb{R}^2 \qquad (i > m_n),$$

where the coordinates are independent, centered, L^2-normalized, and satisfy $\|\xi_{n,i}\|_{L^1} \geq \delta$. Set

$$\alpha_{n,i} := \big((a_i^{(n)})^2 + (b_i^{(n)})^2\big)^{1/2}.$$

Assume

$$\max_{i>m_n} \alpha_{n,i} \to 0, \qquad \sum_{i>m_n} \alpha_{n,i}^2 \to \sigma^2 > 0,$$

and that the row sums $\sum_{i>m_n} V_{n,i}$ converge in distribution to an infinitely divisible law μ on $\mathbb{R}^2$. Then the Gaussian part of μ is nontrivial.

Proof. Set $\beta_n = \max_{i>m_n} \alpha_{n,i}$ and $r_n = \beta_n^{1/2}$. For each $i > m_n$ with $\alpha_{n,i} \neq 0$, let $K_{n,i} = r_n/\alpha_{n,i} \to \infty$ uniformly and define the centered truncation

$$\widetilde{\xi}_{n,i} := \xi_{n,i}\mathbf{1}_{\{|\xi_{n,i}|\leq K_{n,i}\}} - \mathbb{E}[\xi_{n,i}\mathbf{1}_{\{|\xi_{n,i}|\leq K_{n,i}\}}].$$

Since $\mathbb{E}[|\xi_{n,i}|] \geq \delta$ and $\mathbb{E}[\xi_{n,i}^2] = 1$, the same estimate as in the complex proof shows that there exists $c_0 = c_0(\delta) > 0$ such that

$$\mathbb{E}[\widetilde{\xi}_{n,i}^2] \geq c_0$$

for all large n and all $i > m_n$. Define $\widetilde{V}_{n,i} = (a_i^{(n)}\widetilde{\xi}_{n,i},\, b_i^{(n)}\widetilde{\xi}_{n,i})$. Then $|\widetilde{V}_{n,i}| \leq 2r_n$, so the truncated array is bounded and infinitesimal, and

$$\sum_{i>m_n} \mathbb{E}[|\widetilde{V}_{n,i}|^2] \geq c_0 \sum_{i>m_n} \alpha_{n,i}^2 \longrightarrow c_0\sigma^2 > 0.$$

For infinitesimal triangular arrays, the covariance matrices of shrinking centered truncations converge to the Gaussian covariance of the limit law; see [33, Chapter IV]. Since the traces of the truncation covariances stay bounded away from zero, the Gaussian covariance of μ is nonzero. □

Remark 3.8 (Alternative Gaussian endpoint)*.* Assume we are in the degenerate-head regime $H_X \equiv H_Y \equiv 0$, so the tail law μ of (R_X, R_Y) is supported on $\Gamma = \{(u,v) : |u| = |v|\}$. Write $\mu = \gamma * \nu$, where γ is the Gaussian part and ν the independent remainder. By Proposition 3.7, γ is nontrivial. Its support is a nonzero linear subspace $E \subset \mathbb{R}^2$.

Because $\operatorname{supp}\mu = \operatorname{supp}\gamma + \operatorname{supp}\nu \subset \Gamma = L_+ \cup L_-$, the connected set E must itself lie in one of the lines L_τ. Fix such a τ. For any $z \in \operatorname{supp}\nu$, the translated line $E + z$ is contained in Γ. Since a translate of L_τ not equal to L_τ intersects $L_+ \cup L_-$ in at most two points, one must have $z \in L_\tau$. Hence $\operatorname{supp}\nu \subset L_\tau$ and therefore $\operatorname{supp}\mu \subset L_\tau$. Thus $R_Y = \tau R_X$ almost surely. This recovers the final contradiction without appealing to Lemma 3.6.

4. Second Proof: Anticoncentration

This section gives a second proof of Theorem 1.1. The argument is quantitative in spirit. Instead of passing to a limit law, we split the coefficients into a head and a tail and show directly that one of two explicit mechanisms forces a positive modulus gap.

By Proposition 2.2, it is enough to prove an orthogonal lower bound for arbitrary unit vectors in the one-exception span. Thus let $a = (a_i)_{i\geq 1}, b = (b_i)_{i\geq 1} \in \ell^2(\mathbb{N})$ satisfy

$$\|a\|_2 = \|b\|_2 = 1, \qquad \langle a, b\rangle = 0,$$

and set

$$X := a_1\eta + \sum_{i\geq 2} a_i\xi_i, \qquad Y := b_1\eta + \sum_{i\geq 2} b_i\xi_i.$$

Because the family $(\eta, \xi_i)_{i\geq 2}$ is orthonormal in L^2, these sums converge in L^2. We shall prove that

$$\||X| - |Y|\|_{L^2} \geq c_{\delta,\eta} > 0,$$

where the constant depends only on δ and on the law of η.

4.1. **Setup and the dichotomy.** Define

$$\varepsilon := \mathbb{E}[|X^2 - Y^2|], \qquad \Delta := \||X| - |Y|\|_{L^2}.$$

As before,

$$\varepsilon \leq 2\Delta.$$

Fix

$$\theta := \frac{\delta^2}{8}, \qquad \lambda := \frac{\delta^3}{2^{20}}, \qquad s_0 := \frac{\delta^4}{2^{20}}.$$

Let S_η be the bounded linear probe from Proposition 3.3. If $\eta^2 \not\equiv 1$, let T_η be the bounded quadratic probe from the same proposition. Choose a constant $K_{\delta,\eta} \geq 1$ so large that

$$K_{\delta,\eta} \geq \frac{4}{\delta^2}, \qquad K_{\delta,\eta} \geq \frac{2}{\delta}, \qquad K_{\delta,\eta} \geq \|S_\eta\|_{L^\infty},$$

and, if $\eta^2 \not\equiv 1$,

$$K_{\delta,\eta} \geq \|T_\eta\|_{L^\infty}.$$

Define the head set by

$$H = H(a, b) := \{1\} \cup \{i \geq 2 : \max(|a_i|, |b_i|) > \lambda\}.$$

Then

$$\#H \leq d^*(\delta) := 3 + 2^{41}\delta^{-6},$$

and every tail coordinate $i \notin H$ satisfies

$$\max(|a_i|, |b_i|) \leq \lambda.$$

The first step is to separate the proof into two mutually exclusive regimes. Either one of the two signs makes the tail nearly one-dimensional, in which case the argument is driven by head rigidity, or both signs retain substantial tail mass, in which case anticoncentration becomes effective.

Lemma 4.1 (Tail sign dichotomy)**.** *Exactly one of the following alternatives holds:*

(i) ***Diffuse:***

$$\|(a - b)\mathbf{1}_{H^c}\|_2^2 \geq 16s_0 \quad \textit{and} \quad \|(a + b)\mathbf{1}_{H^c}\|_2^2 \geq 16s_0.$$

(ii) ***Concentrated:*** *there exists $\tau \in \{\pm 1\}$ such that*

$$\|(a - \tau b)\mathbf{1}_{H^c}\|_2 < \theta/32.$$

Proof. We first note that

$$16s_0 = \left(\frac{\theta}{32}\right)^2.$$

Thus the concentrated alternative may be rewritten as saying that there exists $\tau \in \{\pm 1\}$ such that

$$\|(a - \tau b)\mathbf{1}_{H^c}\|_2^2 < 16s_0.$$

If this happens for $\tau = 1$ or $\tau = -1$, then we are in the concentrated case and there is nothing more to prove. Suppose, on the contrary, that the concentrated alternative fails. Then for both choices of sign we have

$$\|(a - b)\mathbf{1}_{H^c}\|_2^2 \geq 16s_0 \qquad \text{and} \qquad \|(a + b)\mathbf{1}_{H^c}\|_2^2 \geq 16s_0.$$

But this is exactly the diffuse alternative.

Finally, the two alternatives are mutually exclusive, because if the concentrated alternative held for some sign τ, then one of the two vectors $a - b$ or $a + b$ would have tail norm squared strictly smaller than $16s_0$, which contradicts the diffuse inequalities. Hence exactly one of the two alternatives holds. □

4.2. **The concentrated branch.** We now turn to the first regime. In the concentrated case, the tail of $a - \tau b$ is small for a suitable sign τ, so any substantial discrepancy between the two vectors must be visible on the head. The next proposition shows that the quadratic observable $X^2 - Y^2$ controls precisely those head coefficients.

Proposition 4.2 (Head entry bounds with one exceptional coordinate)**.** *The following estimates hold:*

(1) *if $i, j \in H \cap \{2, 3, \dots\}$, then*

$$|a_i a_j - b_i b_j| \leq K_{\delta,\eta}\, \varepsilon;$$

(2) *if $j \in H \cap \{2, 3, \dots\}$, then*

$$|a_1 a_j - b_1 b_j| \leq K_{\delta,\eta}\, \varepsilon;$$

(3) *if $\eta^2 \not\equiv 1$, then*

$$|a_1^2 - b_1^2| \leq K_{\delta,\eta}\, \varepsilon.$$

Proof. For $i = j \geq 2$, use the quadratic probe T_i for ξ_i:

$$|a_i^2 - b_i^2| = \left|\mathbb{E}[(X^2 - Y^2)T_i]\right| \leq \|T_i\|_{L^\infty}\varepsilon \leq K_{\delta,\eta}\varepsilon.$$

For distinct $i, j \geq 2$, use the linear probes S_i, S_j:

$$2|a_i a_j - b_i b_j| = \left|\mathbb{E}[(X^2 - Y^2)S_i S_j]\right| \leq \|S_i\|_{L^\infty}\|S_j\|_{L^\infty}\varepsilon \leq K_{\delta,\eta}\varepsilon.$$

This proves (1).

For (2), use the product $S_\eta S_j$. Exactly as in the proof of Lemma 3.4, all terms vanish except the mixed exceptional-good term, so

$$2|a_1 a_j - b_1 b_j| = \left|\mathbb{E}[(X^2 - Y^2)S_\eta S_j]\right| \leq \|S_\eta\|_{L^\infty}\|S_j\|_{L^\infty}\varepsilon \leq K_{\delta,\eta}\varepsilon.$$

Finally, if $\eta^2 \not\equiv 1$, use T_η:

$$|a_1^2 - b_1^2| = \left|\mathbb{E}[(X^2 - Y^2)T_\eta]\right| \le \|T_\eta\|_{L^\infty}\varepsilon \le K_{\delta,\eta}\varepsilon.$$

□

To convert these entrywise bounds into a global contradiction, we use the same rank-one algebra that appeared in the homogeneous proof. This isolates the geometric fact that two orthogonal unit vectors cannot agree too well on the head unless the corresponding head difference matrix is small.

Lemma 4.3 (Rank-one algebra)**.** *For $x, y \in \mathbb{R}^m$, let $D(x,y)_{ij} = x_i x_j - y_i y_j$. Then*

$$\|x-y\|_2^2\,\|x+y\|_2^2 \le 2\|D(x,y)\|_F^2.$$

Proof. Writing $s = \|x\|_2^2$, $t = \|y\|_2^2$, and $c = \langle x, y\rangle$, the left-hand side equals $(s+t)^2 - 4c^2$, while the right-hand side is $2(s^2 + t^2 - 2c^2)$. Their difference is $(s-t)^2$, which is nonnegative. □

Combining the head-entry bounds with the rank-one inequality yields the concentrated branch. The proof splits off the genuinely degenerate Rademacher-only head, since only there can the exceptional coordinate fail to provide a quadratic probe.

Proposition 4.4 (Concentrated branch)**.** *If there exists $\tau \in \{\pm 1\}$ such that*

$$\|(a - \tau b)\mathbf{1}_{H^c}\|_2 \le \theta/32,$$

then

$$\varepsilon \ge c_{\text{line}}(\delta, \eta) > 0.$$

Proof. We distinguish two cases.

Case 1: η is not Rademacher, or $H \cap \{2, 3, \dots\}$ is nonempty. In this case the head coefficients are rigidly linked by Proposition 4.2: either directly through the diagonal estimate for the exceptional coordinate or, in the Rademacher case, through a good head coordinate exactly as in the proof of Lemma 3.4. In particular there exists a sign, which must be the same τ, such that the full head difference matrix of $a\mathbf{1}_H$ and $\tau b\mathbf{1}_H$ has all entries bounded by $K_{\delta,\eta}\varepsilon$.

Let

$$x := a\mathbf{1}_H, \qquad y := \tau b\mathbf{1}_H.$$

Applying Lemma 4.3,

$$\|x-y\|_2^2\,\|x+y\|_2^2 \le 2\#H^2 K_{\delta,\eta}^2\varepsilon^2.$$

Since the tail of $a - \tau b$ has norm at most $\theta/32$ and $\|a - \tau b\|_2^2 = 2$ by orthogonality, the head part satisfies

$$\|x-y\|_2^2 = 2 - \|(a-\tau b)\mathbf{1}_{H^c}\|_2^2 \ge 1.$$

Hence

$$\|x+y\|_2 \le 2\#H K_{\delta,\eta}\varepsilon \le 2d^*(\delta)K_{\delta,\eta}\varepsilon.$$

If ε were smaller than a sufficiently small constant $c_{\text{line}}(\delta,\eta)$, then the head of $a + \tau b$ would be tiny while the tail of $a - \tau b$ is already tiny. The same gluing argument as in the homogeneous proof then yields a contradiction. Therefore $\varepsilon \ge c_{\text{line}}(\delta,\eta)$.

Case 2: η is Rademacher and $H = \{1\}$. Write

$$u := a\mathbf{1}_{H^c}, \qquad v := b\mathbf{1}_{H^c}, \qquad \rho := \theta/32,$$

and let

$$U := \sum_{i\geq 2} a_i\xi_i, \qquad V := \sum_{i\geq 2} b_i\xi_i.$$

Since $H = \{1\}$, every coordinate entering U and V is good. The concentrated assumption says precisely that

$$\|u - \tau v\|_2 = \|U - \tau V\|_{L^2} \leq \rho.$$

Set

$$R := U - \tau V, \qquad S := U + \tau V, \qquad \alpha := a_1 - \tau b_1, \qquad \beta := a_1 + \tau b_1.$$

Then

$$X - \tau Y = \alpha\eta + R, \qquad X + \tau Y = \beta\eta + S.$$

Because $\|a\|_2 = \|b\|_2 = 1$ and $\langle a, b\rangle = 0$, we have

$$\|a - \tau b\|_2^2 = \|a + \tau b\|_2^2 = 2.$$

Therefore

$$\alpha^2 + \|R\|_{L^2}^2 = \alpha^2 + \|u - \tau v\|_2^2 = 2, \qquad \beta^2 + \|S\|_{L^2}^2 = \beta^2 + \|u + \tau v\|_2^2 = 2.$$

In particular,

$$|\alpha| \geq \sqrt{2 - \rho^2} \geq 1.$$

We next bound β. Since

$$|\alpha\beta| = |a_1^2 - b_1^2| = \big|\|v\|_2^2 - \|u\|_2^2\big| = \big|\langle u - \tau v, u + \tau v, \rangle\big|,$$

it follows that

$$|\alpha\beta| \leq \|u - \tau v\|_2\|u + \tau v\|_2 = \|R\|_{L^2}\|S\|_{L^2} \leq \sqrt{2}\,\rho.$$

Using $|\alpha| \geq 1$, we obtain

$$|\beta| \leq \sqrt{2}\,\rho.$$

Hence $\|S\|_{L^2} \geq 1$, and Lemma 2.5 gives

$$\|S\|_{L^1} = \Big\|\sum_{i\geq 2}(a_i + \tau b_i)\xi_i\Big\|_{L^1} \geq \frac{\delta}{2\sqrt{2}}\|u + \tau v\|_2 \geq \frac{\delta}{2\sqrt{2}}.$$

On the other hand,

$$\|R\|_{L^1} \leq \|R\|_{L^2} \leq \rho.$$

Now condition on the sigma-algebra generated by the good coordinates. Since η is an independent Rademacher variable,

$$X^2 - Y^2 = (X - \tau Y)(X + \tau Y) = (\alpha\eta + R)(\beta\eta + S) = \alpha\beta + RS + \eta(\alpha S + \beta R).$$

Therefore

$$\begin{aligned}\mathbb{E}\big[|X^2 - Y^2| \mid R, S\big] &= \frac{1}{2}\Big(|\alpha\beta + RS + \alpha S + \beta R| + |\alpha\beta + RS - \alpha S - \beta R|\Big)\\ &= \max\big(|\alpha\beta + RS|, |\alpha S + \beta R|\big) \geq |\alpha S + \beta R|.\end{aligned}$$

Taking expectations and using the triangle inequality, we conclude that

$$\varepsilon \geq |\alpha|\|S\|_{L^1} - |\beta|\|R\|_{L^1} \geq \frac{\delta}{2\sqrt{2}} - \sqrt{2}\,\rho^2.$$

Since $\rho = \theta/32 = \delta^2/256$ and $0 < \delta \leq 1$, the right-hand side is bounded below by a positive constant depending only on δ (for instance by $\delta/4$). After decreasing $c_{\mathrm{line}}(\delta, \eta)$ if necessary, this proves Case 2. □

4.3. **The diffuse branch.** We now move to the complementary regime. Here both $a-b$ and $a+b$ retain a definite amount of ℓ^2 mass on the tail, and every tail coordinate is individually small. This is exactly the setting in which block anticoncentration can be used.

Set

$$c := \frac{(a-b)\mathbf{1}_{H^c}}{\sqrt{2}}, \qquad d := \frac{(a+b)\mathbf{1}_{H^c}}{\sqrt{2}}.$$

Because $1 \in H$, the vectors c and d are supported only on good coordinates. In the diffuse case both have ℓ^2 norm at least $\sqrt{8s_0}$, while each coordinate is bounded by $\sqrt{2}\,\lambda$.

The next lemma packages the quantitative probabilistic input. It says that after arbitrary deterministic shifts, the two tail combinations cannot both remain close to zero with high probability.

Lemma 4.5 (Quantitative cross anticoncentration)**.** *There exists $r_* = r_*(\delta) > 0$ such that, in the diffuse case,*

$$\mathbb{P}\big(r_* < \min(|X_c + u|, |X_d + v|)\big) \geq \frac{1}{4} \qquad (u, v \in \mathbb{R}).$$

Proof. We prove first a one-variable small-ball estimate. Let $e \in \ell^2(\mathbb{N})$ be supported on H^c and satisfy

$$\|e\|_2^2 \geq 8s_0, \qquad \sup_{i\geq 2}|e_i| \leq \sqrt{2}\,\lambda.$$

We claim that there exists $r_0 = r_0(\delta) > 0$ such that

$$\mathbb{P}\big(|X_e - t| \leq r_0\big) \leq \frac{3}{8} \qquad (t \in \mathbb{R}). \tag{4.1}$$

Once this is proved, the lemma follows immediately by applying (4.1) to $e = c$ with $t = -u$ and to $e = d$ with $t = -v$. Indeed, in the diffuse case both c and d satisfy the displayed hypotheses, so

$$\mathbb{P}\big(|X_c + u| \leq r_0\big) \leq \frac{3}{8}, \qquad \mathbb{P}\big(|X_d + v| \leq r_0\big) \leq \frac{3}{8}.$$

By the union bound,

$$\mathbb{P}\big(|X_c + u| \leq r_0 \text{ or } |X_d + v| \leq r_0\big) \leq \frac{3}{4},$$

and therefore

$$\mathbb{P}\big(r_0 < \min(|X_c + u|, |X_d + v|)\big) \geq \frac{1}{4}.$$

Thus it remains only to establish (4.1).

Step 1: a block partition. Set

$$M := \left\lceil 2^{16}\delta^{-2} \right\rceil, \qquad \eta := \frac{s_0}{2M}, \qquad r_0 := \frac{\delta\sqrt{\eta}}{16}.$$

Because

$$2\lambda^2 = \frac{\delta^6}{2^{39}} \leq \frac{\delta^6}{2^{38}} \leq \eta,$$

every coordinate satisfies $e_i^2 \le \eta$. Since

$$\|e\|_2^2 \ge 8s_0 > 2M\eta = s_0,$$

we may choose pairwise disjoint finite sets $G_1, \dots, G_M \subset H^c$ such that

$$\eta \le \sum_{i \in G_j} e_i^2 \le 2\eta \qquad (1 \le j \le M). \tag{4.2}$$

Indeed, construct the blocks greedily. Suppose $G_1, \dots, G_{j-1}$ have already been chosen. Their total energy is at most $2(j-1)\eta$, so the unused part of e still has energy at least

$$8s_0 - 2(j-1)\eta \ge 8s_0 - 2M\eta = 7s_0 > \eta.$$

Hence one can choose a finite subset of the remaining support whose energy is at least η, and by minimality its energy is at most $\eta + \sup_i e_i^2 \le 2\eta$. This produces the next block.

Step 2: symmetrized block sums have uniform two-sided mass. Let X_e' be an independent copy of X_e, realized on a second copy of the probability space. For each $i \ge 2$ define

$$D_i := \xi_i(\omega_1) - \xi_i(\omega_2).$$

Then the variables $(D_i)_{i \ge 2}$ are independent, centered, and symmetric, with

$$\|D_i\|_{L^2}^2 = \mathbb{E}[(\xi_i - \xi_i')^2] = 2.$$

Moreover, by Jensen's inequality applied conditionally on ξ_i,

$$\mathbb{E}[|D_i|] = \mathbb{E}\big[\mathbb{E}[|\xi_i - \xi_i'| \mid \xi_i]\big] \ge \mathbb{E}\big[|\xi_i - \mathbb{E}[\xi_i']|\big] = \mathbb{E}[|\xi_i|] \ge \delta.$$

Hence

$$\mathbb{E}[|D_i|] \ge \frac{\delta}{\sqrt{2}} \|D_i\|_{L^2}.$$

For each block define

$$Y_j := \sum_{i \in G_j} e_i D_i, \qquad \sigma_j^2 := \sum_{i \in G_j} e_i^2.$$

By (4.2),

$$\eta \le \sigma_j^2 \le 2\eta.$$

Applying Lemma 2.5 to the independent centered family $(D_i)_{i \in G_j}$, with lower L^1 ratio $\delta/\sqrt{2}$, gives

$$\mathbb{E}[|Y_j|] \ge \frac{(\delta/\sqrt{2})}{2\sqrt{2}} \Big(\sum_{i \in G_j} e_i^2 \|D_i\|_{L^2}^2 \Big)^{1/2} = \frac{\delta}{4} \big(2\sigma_j^2\big)^{1/2} = \frac{\delta}{2}\sigma_j \ge \frac{\delta\sqrt{\eta}}{2}.$$

Set

$$a_0 := \frac{\delta\sqrt{\eta}}{4} = 4r_0.$$

Then $a_0 \le \mathbb{E}[|Y_j|]/2$. Let

$$A_j := \{|Y_j| \ge a_0\}.$$

Since

$$\mathbb{E}[Y_j^2] = 2\sigma_j^2 \le 4\eta,$$

we obtain

$$\mathbb{E}[|Y_j|] \le a_0 + \mathbb{E}[|Y_j|\mathbf{1}_{A_j}] \le a_0 + \mathbb{P}(A_j)^{1/2}\mathbb{E}[Y_j^2]^{1/2}.$$

Because $\mathbb{E}[|Y_j|] \ge 2a_0$, the second term on the right is at least a_0, so

$$\mathbb{P}(A_j)^{1/2}\,(4\eta)^{1/2} \ge a_0.$$

Therefore

$$\mathbb{P}(A_j) \ge \frac{a_0^2}{4\eta} = \frac{\delta^2}{64}. \tag{4.3}$$

Step 3: many active blocks with high probability. Let

$$N := \sum_{j=1}^{M} \mathbf{1}_{A_j}.$$

The variables $\mathbf{1}_{A_j}$ are independent because the blocks are disjoint. By (4.3),

$$\mathbb{E}[N] = \sum_{j=1}^{M} \mathbb{P}(A_j) \ge M\frac{\delta^2}{64} \ge 1024.$$

Since $\mathrm{Var}(\mathbf{1}_{A_j}) \le \mathbb{E}[\mathbf{1}_{A_j}]$, we also have

$$\mathrm{Var}(N) \le \mathbb{E}[N].$$

Chebyshev's inequality gives

$$\mathbb{P}(N < 256) \le \mathbb{P}\big(|N - \mathbb{E}[N]| \ge \mathbb{E}[N] - 256\big) \le \frac{\mathrm{Var}(N)}{(\mathbb{E}[N]-256)^2} \le \frac{\mathbb{E}[N]}{(\mathbb{E}[N]-256)^2} \le \frac{1024}{768^2} < \frac{1}{16}.$$

Step 4: a Sperner bound for the symmetrized sum. Let

$$U := \sum_{j=1}^{M} Y_j.$$

We claim that for every $s \in \mathbb{R}$,

$$\mathbb{P}(|U - s| \le 2r_0) \le \frac{1}{8}. \tag{4.4}$$

Indeed,

$$\mathbb{P}(|U-s| \le 2r_0) \le \mathbb{P}(|U-s| \le 2r_0,\ N \ge 256) + \mathbb{P}(N < 256).$$

We have already bounded the second term by $1/16$, so it remains to estimate the first one.

Condition on the absolute values $(|Y_j|)_{j=1}^M$. Since each Y_j is symmetric, conditional on $(|Y_j|)_j$ the signs of the nonzero Y_j are independent Rademacher variables. On the event $\{N \ge 256\}$ let

$$J := \{j : |Y_j| \ge a_0\}, \qquad m := \#J \ge 256.$$

After freezing the inactive coordinates and the magnitudes $|Y_j|$, the condition $|U-s| \le 2r_0$ becomes

$$\sum_{j\in J} \varepsilon_j w_j \in I,$$

where $w_j := |Y_j| \ge a_0$ and I is an interval of diameter $4r_0 = a_0$. Since $a_0 < 2w_j$ for every $j \in J$, the family of sign patterns for which the weighted sum falls in I is an antichain in $\{\pm 1\}^J$. Indeed,

if two such sign patterns were comparable in the natural subset order, their sums would differ by at least $2w_j \geq 2a_0 > a_0$, which is impossible inside an interval of diameter a_0. By Sperner's theorem, the conditional probability of this event is therefore at most

$$\frac{\binom{m}{\lfloor m/2 \rfloor}}{2^m} \leq \frac{1}{\sqrt{m}} \leq \frac{1}{16},$$

because $m \geq 256$. Averaging over the conditioning yields

$$\mathbb{P}(|U - s| \leq 2r_0,\ N \geq 256) \leq \frac{1}{16}.$$

Together with the bound for $\mathbb{P}(N < 256)$, this proves (4.4).

Step 5: unsymmetrization. Let

$$R := \sum_{i \notin G_1 \cup \cdots \cup G_M} e_i D_i.$$

Then R is independent of U, since it is built from coordinates disjoint from those entering the block sums. Also,

$$X_e(\omega_1) - X_e(\omega_2) = U + R.$$

Fix $t \in \mathbb{R}$. If both $|X_e(\omega_1) - t| \leq r_0$ and $|X_e(\omega_2) - t| \leq r_0$, then

$$|X_e(\omega_1) - X_e(\omega_2)| \leq 2r_0.$$

Hence

$$\mathbb{P}(|X_e - t| \leq r_0)^2 \leq \mathbb{P}(|X_e(\omega_1) - X_e(\omega_2)| \leq 2r_0) = \mathbb{P}(|U + R| \leq 2r_0).$$

Conditioning on R and using (4.4),

$$\mathbb{P}(|U + R| \leq 2r_0) \leq \sup_{s \in \mathbb{R}} \mathbb{P}(|U - s| \leq 2r_0) \leq \frac{1}{8}.$$

Therefore

$$\mathbb{P}(|X_e - t| \leq r_0) \leq \frac{1}{\sqrt{8}} < \frac{3}{8}.$$

This proves (4.1), and with it the lemma. □

Once this shifted anticoncentration estimate is available, the lower bound for ε follows from the same factorization of $X^2 - Y^2$ that was already used in the compactness proof.

Proposition 4.6 (Diffuse branch)**.** *If the diffuse alternative of Lemma 4.1 holds, then*

$$\varepsilon \geq c_{\text{cross}}(\delta) > 0.$$

Proof. Let

$$h_\pm := X_{(a \pm b)\mathbf{1}_H}.$$

Then

$$X^2 - Y^2 = 2\big(X_d + h_+/\sqrt{2}\big)\big(X_c + h_-/\sqrt{2}\big),$$

and therefore

$$|X^2 - Y^2| \geq 2 \min\Big(|X_d + h_+/\sqrt{2}|, |X_c + h_-/\sqrt{2}|\Big)^2.$$

Condition on the head variables $h_\pm$ and apply Lemma 4.5 with the shifts $u = h_-/\sqrt{2}$ and $v = h_+/\sqrt{2}$. This yields a uniform positive lower bound for the conditional expectation of the squared minimum, hence for ε. The resulting constant depends only on δ. □

4.4. **Conclusion of the second proof.** At this point the proof is immediate: the dichotomy shows that every orthogonal pair falls into exactly one of the two branches, and each branch gives a quantitative lower bound for the same quantity ε.

Second proof of Theorem 1.1. By Proposition 2.2, it is enough to prove a uniform orthogonal lower bound. Let a, b be orthogonal unit vectors and form X and Y as above. Apply the dichotomy of Lemma 4.1.

If the concentrated alternative holds, then Proposition 4.4 yields

$$\varepsilon \geq c_{\text{line}}(\delta, \eta).$$

If the diffuse alternative holds, then Proposition 4.6 yields

$$\varepsilon \geq c_{\text{cross}}(\delta).$$

Since $\varepsilon \leq 2\Delta$, we conclude that

$$\Delta = \||X| - |Y|\|_{L^2} \geq \frac{1}{2} \min\bigl(c_{\text{line}}(\delta, \eta), c_{\text{cross}}(\delta)\bigr) =: c_{\delta,\eta} > 0.$$

The orthogonal reduction then gives

$$\min_{\sigma \in \{\pm 1\}} \|X - \sigma Y\|_{L^2} \leq \frac{\sqrt{2}}{c_{\delta,\eta}} \||X| - |Y|\|_{L^2}$$

for all $X, Y \in \overline{\text{span}}(\eta, \xi_i : i \geq 2)$, completing the proof of Theorem 1.1. □

Corollary 4.7 (Finite-dimensional one-exception stable phase retrieval)**.** *Let $n \geq 2$. Then there exists $C_{\delta,\eta} \geq 1$ such that, for all*

$$X, Y \in \text{span}(\eta, \xi_2, \ldots, \xi_n),$$

one has

$$\min_{\sigma \in \{\pm 1\}} \|X - \sigma Y\|_{L^2} \leq C_{\delta,\eta} \||X| - |Y|\|_{L^2}.$$

Proof. This is an immediate consequence of Theorem 1.1, since

$$\text{span}(\eta, \xi_2, \ldots, \xi_n) \subset \overline{\text{span}}(\eta, \xi_i : i \geq 2).$$

□

5. Comments and Open Problems

The compactness and quantitative proofs suggest several natural directions for further work. Although the theorem proved here settles the qualitative one-exception problem in the real Hilbertian setting, the mechanisms that enter the proof are flexible enough that one is led almost immediately to adjacent questions. Some of these concern extensions of the model, while others concern the sharp quantitative form of the present result. We briefly record several of them here.

Remark 5.1 (Complex-valued models). The complex one-exception theorem already exhibits the same head–tail architecture, but the endpoint geometry is richer because the analogue of the phase-retrieval cone is foliated by complex lines and conjugate planes rather than by two real lines. The role of the upper L^1 bound is then supplemented by a generalized-circle gap. The real argument presented here should be viewed as the cleanest model case for that geometry. This makes the complex setting an especially interesting next test case: it is close enough to the present problem that one expects the same structural ideas to survive, but different enough that the endpoint classification becomes genuinely more delicate.

Remark 5.2 (More general symmetries). Classical phase retrieval is recovery modulo the group $\{\pm 1\}$. It is natural to ask for analogous theorems modulo larger compact symmetry groups or for quotient metrics arising from finite reflection groups. The compactness proof is particularly well suited to that direction because it separates the problem into an exact limit identity and a geometric classification of the corresponding support set. For that reason, this is not merely a formal generalization: it points toward a broader geometric framework in which the support theorem, rather than the specific real-sign symmetry, is the central object.

The next questions concern the extent to which the present theorem can be sharpened. The first asks for a genuinely effective version of the one-exception theorem itself.

Question 5.3. Can one obtain a fully quantitative version of Theorem 1.1 with one exceptional coordinate, with constants depending explicitly on δ and quantitative parameters of the exceptional variable η?

This question is particularly interesting because the two proofs in the paper point in different directions: the compactness proof is robust but soft, while the second proof isolates several places where explicit constants might, at least in principle, be tracked.

Closely related to that issue is the problem of identifying the correct endpoint assumption on the good coordinates.

Question 5.4. What is the optimal endpoint replacement for the upper bound $\|\xi_i\|_{L^1} \leq 1 - \delta$? In the real case this bound excludes the Rademacher obstruction, but it is not obvious whether a weaker quantitative non-flatness parameter could suffice.

This is again an interesting structural question rather than a matter of constant optimization. The present upper bound is easy to interpret and natural for the proof, but it would be valuable to know whether the real obstruction is exactly Rademacher degeneracy or some larger non-flatness phenomenon.

Finally, one may ask how much of the theorem belongs specifically to Hilbert space geometry, and how much should persist in a wider Banach-function-space setting.

Question 5.5. Which Banach-function-space analogues of the present theorem remain true once the ambient space is no longer Hilbertian? The orthogonal reduction is special to Hilbert spaces, so a different mechanism would be needed.

This problem seems especially compelling because the function-space formulation of stable phase retrieval is not inherently Hilbertian, even though the present proof is. Any satisfactory extension

would likely require replacing orthogonality by a more flexible geometric notion while retaining enough structure to control the modulus map.

Remark 5.6. The quantitative proof is stronger whenever one seeks effective constants, but the compactness proof is likely the more durable template for extensions. It uses only bounded probes, a profile decomposition, and a structural description of diffuse limits. Those ingredients are robust enough to adapt to complex models, to settings with exceptional coordinates, and plausibly to retrieval modulo more complicated symmetries. Taken together, these questions suggest that the present theorem is less an endpoint than a model case: it captures a setting in which the geometry can still be classified cleanly, while already displaying most of the phenomena that one would expect to confront in more general phase-retrieval problems.

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